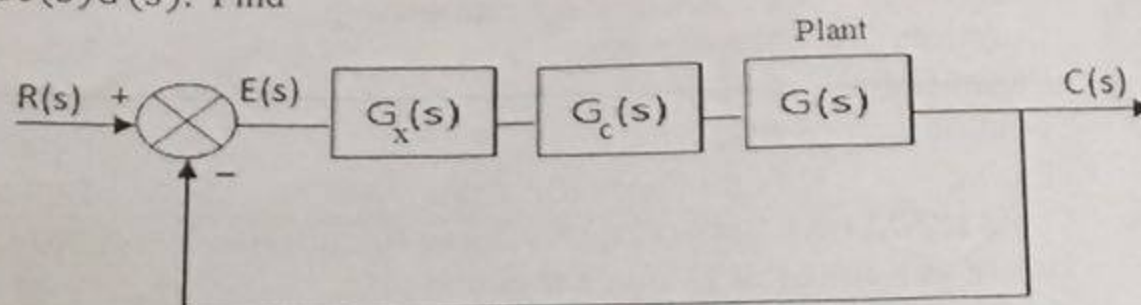
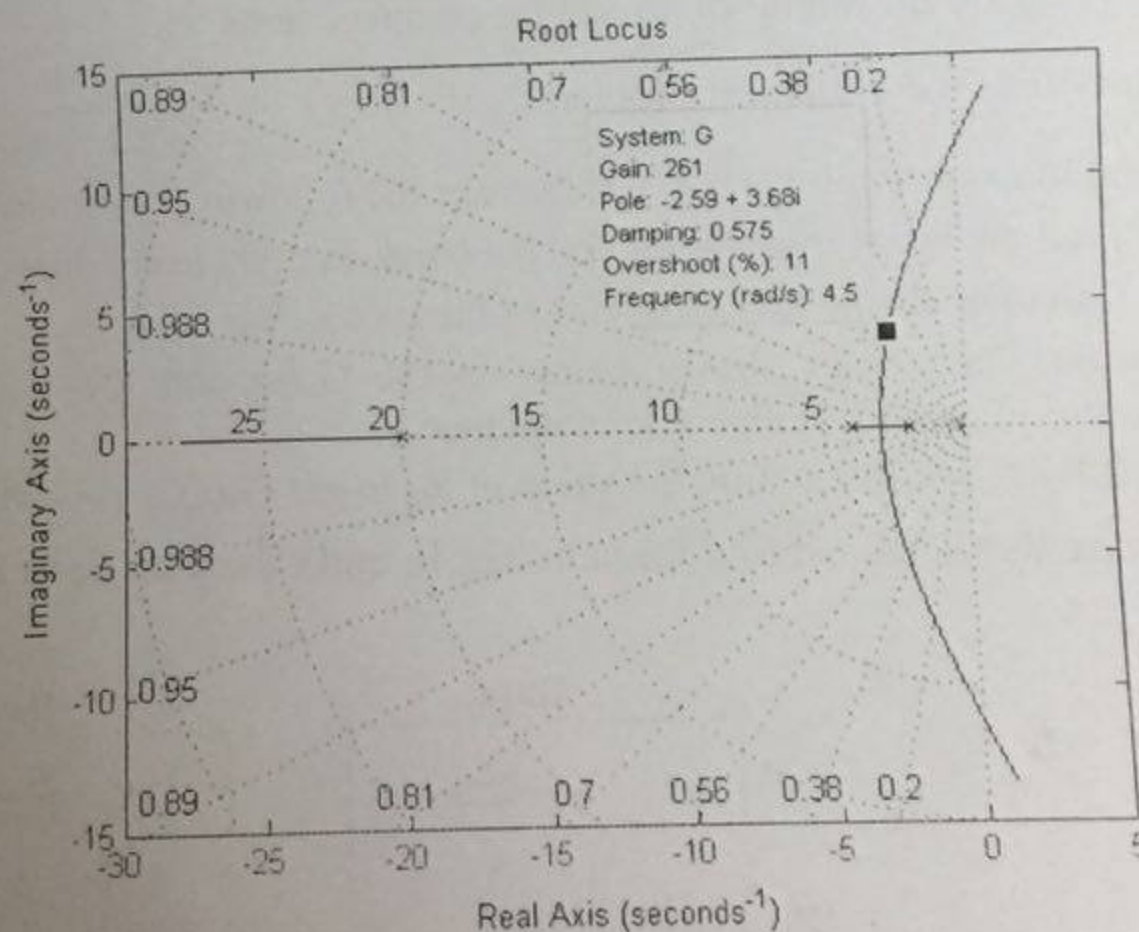


5. (10 points) The compensator circuit of question 4 is connected in cascade with a plant having $G(s) = \frac{1}{(s+1)(s+2)}$, as shown below. The forward path transfer function is given by $F(s) = G_x(s) G_c(s) G(s)$. Find



- (a) the three static error constants K_p , K_v , and K_a for the compensated system.
 (b) the steady state error for a unit step input, a unit ramp input and a unit parabola input after compensation.
6. (10 points) Given $G(s) = \frac{K}{(s^2+2s+2)}$, find the magnitude M of $G(j\omega)$ in decibels, and phase P of $G(j\omega)$ in degrees when $K = 1$ and $\omega = 3$ radians/sec.
7. (10 points) The root locus plot of an uncompensated negative unity feedback control system with an open loop transfer function $G(s) = \frac{K}{(s+2)(s+4)(s+20)}$ is shown below. When $K = 261$, the damping ratio $\zeta = 0.575$, and the dominant poles are at $-2.59 \pm j3.68$, as shown in the root locus plot. It is required to design a LAG compensator for the system, so that the new steady state error for a unit step input is half that of the uncompensated system.
- (a) Find the location of the compensator zero, Z_c , assuming that the location of the compensator pole is at $p_c = -0.001$.
 (b) Also, give the value of the steady state error for the compensated and uncompensated systems. Is the second order approximation valid for the compensated system? Why or why not?



National University of Computer and Emerging Sciences, Lahore Campus



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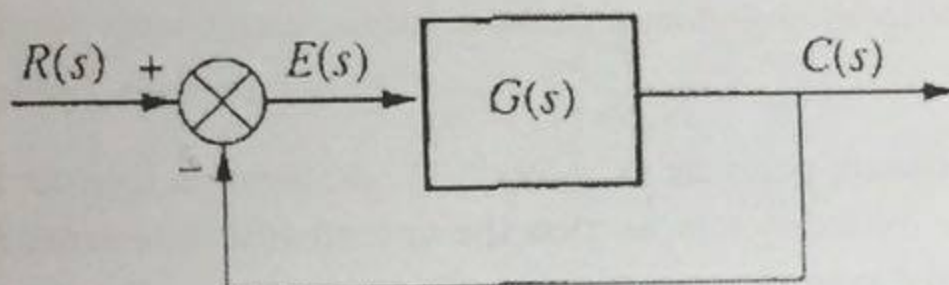
Instruction/Notes: THIS EXAM HAS 10 QUESTIONS. Read ALL questions first. BOX ALL YOUR ANSWERS. Work at a speed of 15 min per question.

1. (10 points) A control system with a closed loop transfer function has a characteristic polynomial given by

$$T(s) = s^5 + 7s^4 + 6s^3 + 42s^2 + 8s + 56$$

Apply the Routh-Hurwitz criterion and find out how many poles of the system are in the right half plane. Is the system stable? Why?

2. (10 points) Consider the unity feedback system shown below:



If $G(s) = K \cdot \frac{s^2 + 2s + 7.5}{s(s+1)(s+2)(s+10)}$ and a root locus is plotted for the closed loop

system for $0 \leq K \leq \infty$, answer the following questions:

- What is the real axis intercept of the asymptotes, and the angles of the asymptotes with the real axis?
- What are the angles of arrival for complex zeros?

3. (10 points) For the system of problem 2, if $G(s) = \frac{K}{(s+2)(s+3)(s+20)}$ and a root locus is plotted for

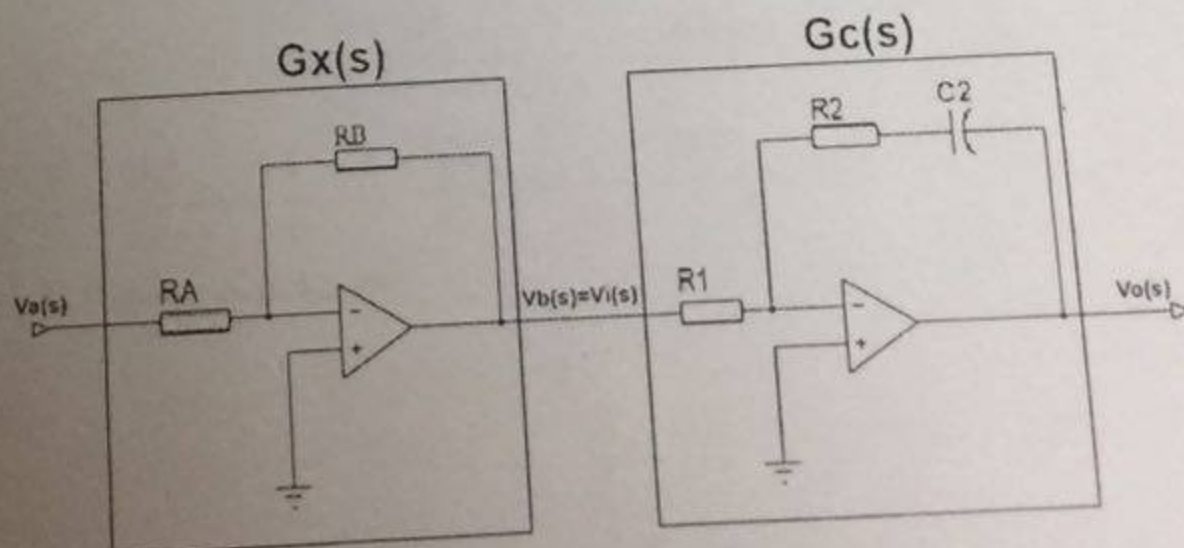
the closed loop system for $0 \leq K \leq \infty$, answer the following questions:

- Find the break-away point and the break-in point using differentiation.
- Calculate the $j\omega$ axis crossings of the root locus.

4. (10 points) The circuit shown below in the $G_x(s)$ and the $G_c(s)$ boxes implements a proportional compensator in cascade with a P.I. compensator.

(a) If $R_B = 10K$ ohms, find the value of R_A to get $G_x(s) = -(100)$.

(b) Use $R_1 = 20K$ and find the values of R_2 and C_2 so that $G_c(s) = -\frac{1}{100} \cdot \frac{s+0.15}{s}$



8. (10 points) For the UNCOMPENSATED system of question # 7, compute the following:

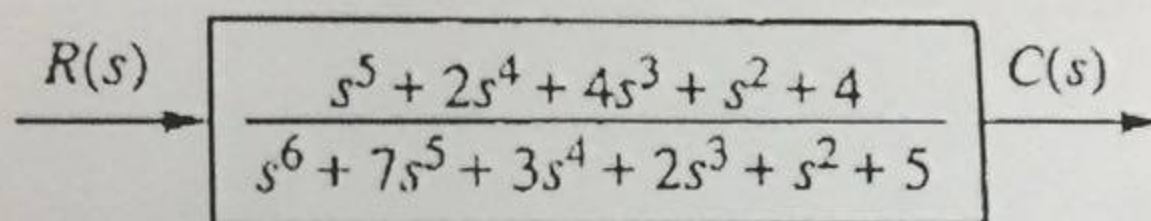
- Settling time, T_s .
- Peak time, T_p .
- Natural frequency, ω_n .
- Damped frequency, ω_d .
- % Overshoot.

9. (10 points) The state-space representation of a control system is given by:

$$\frac{dx}{dt} = Ax + Bu \text{ and } y = Cx + Du. \text{ It is known that } A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{pmatrix}, B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, C = [2 \ 7 \ 1] \text{ and } D=0.$$

Write the transfer function for this system as a ratio of two polynomials.

10. (10 points) Write the differential equation that is mathematically equivalent to the block diagram shown below. Assume $r(t) = 3t^3$.



===== END OF EXAM =====

REMEMBER TO BOX ALL YOUR FINAL ANSWERS.

Some hints:

TABLE 9.7 Types of cascade compensators

Function	Compensator	Transfer function
Improve steady-state error	PI	$K \frac{s + z_c}{s}$
Improve steady-state error	Lag	$K \frac{s + z_c}{s + p_c}$
Improve transient response	PD	$K(s + z_c)$
Improve transient response	Lead	$K \frac{s + z_c}{s + p_c}$
Improve steady-state error and transient response	PID	$K \frac{(s + z_{lag})(s + z_c)}{s}$

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

$$T_s = \frac{4}{\zeta \omega_n}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

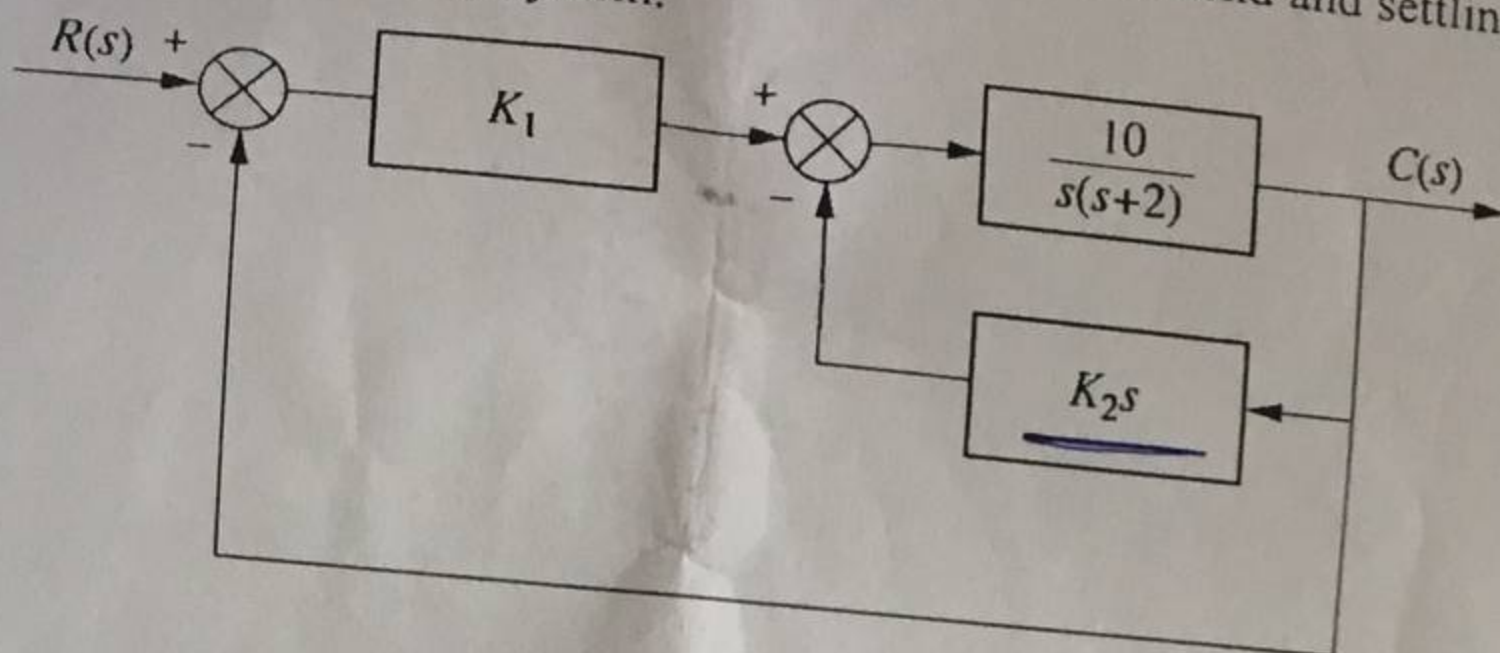
$$\%OS = e^{-(\zeta \pi / \sqrt{1 - \zeta^2})} \times 100$$

$$s_{1,2} = -\zeta \omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

(4.24)



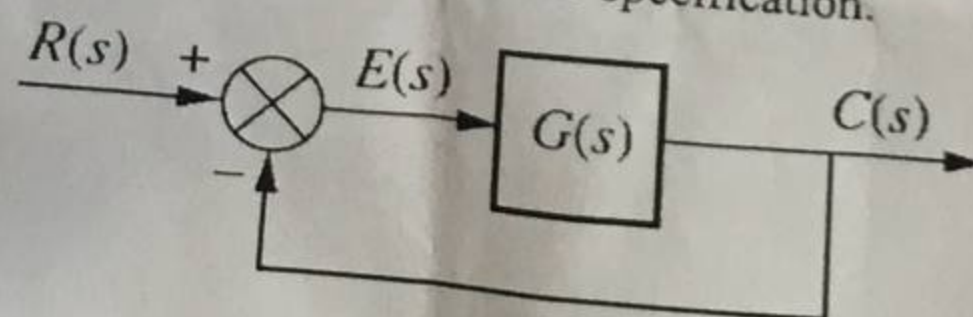
- Q1 Evaluate the value of K_1 and K_2 to yield a peak-time of 1.0 second and settling-time of 2.0 second for the closed-loop system. [10] CLO 3



- Q2 The unity feedback system shown in figure below has an open-loop transfer function $G(s)$, where [10] CLO 5

$$G(s) = \frac{K(s^2 + 3s + 30)}{s^n(s + 5)}$$

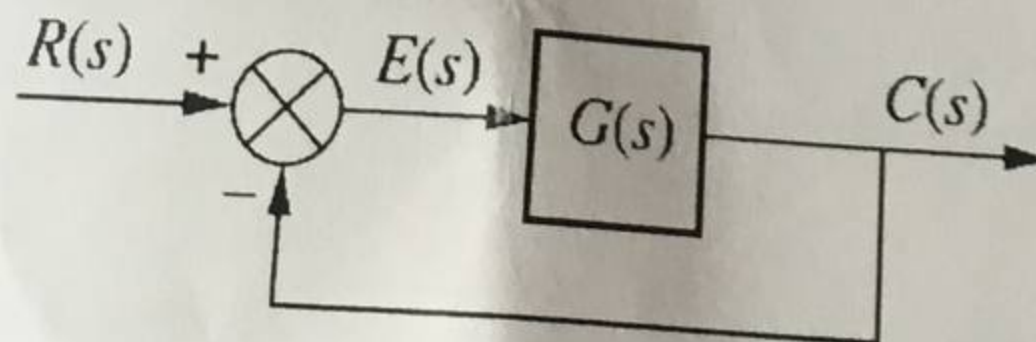
shows a constant steady-state error of $1/6000$ when a ramp input of $10tu(t)$ is applied.
 a) What "Type" of systems exhibit constant steady-state errors for a ramp input.
 b) Hence, find the value of n and K to meet the specification.



- Q3 For the unity feedback system of the following figure, let

$$G(s) = \frac{K(s - 2)(s - 4)}{s^2 + 6s + 25}$$

Clearly sketch the root-locus of the system. Calculate and mention the break-in point and $j\omega$ -crossings on the sketch as well. [10] CLO 4 & 6



$(s+3)$
 $s = -3 \pm j4$